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Inventor(s)	Howard; Damian et al.

Smart nozzle and a surface cleaning device implementing same

Abstract

In general, the present disclosure is directed to nozzle control circuitry for use in surface cleaning devices that preferably reduces overall power consumption of a surface cleaning device by detecting the start of a cleaning operation by a user before energizing one or more components such as an agitator. The nozzle control circuitry can detect a cleaning operation based on data output from one or more sensors (also referred to herein as operation sensors). For example, the nozzle control circuitry can communicate with at least one of a motion sensor such as an accelerometer, an orientation sensor such as gyroscope, and/or an air pressure sensor operatively coupled within a dirty air inlet to detect the presence of generated suction.

Inventors: **Howard; Damian (Winchester, MA), Brown; Andre D. (Natick, MA)**

Applicant: **SharkNinja Operating, LLC** (Needham, MA)

Family ID: **1000008821731**

Assignee: **SharkNinja Operating LLC (Needham, MA)**

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Primary Examiner: Markman; Makena S

Attorney, Agent or Firm: Grossman Tucker Perreault & Pfleger, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application Ser. No. 62/872,862, filed on Jul. 11, 2019, which is fully incorporated herein by reference.

TECHNICAL FIELD

(1) This specification relates to surface cleaning apparatuses, and more particularly, to a surface cleaning device with nozzle control circuitry that can detect usage of the surface cleaning device by a user and activate nozzle components such as a brushroll/agitator, and preferably to adjust brushroll speed, direction of rotation, and/or nozzle orientation relative to a floor type detected proximate the nozzle.

BACKGROUND INFORMATION

(2) Powered surface cleaning devices, such as vacuum cleaners, have multiple components that each receive electrical power from one or more power sources (e.g., one or more batteries or electrical mains). For example, a vacuum cleaner may include a suction motor to generate a vacuum within a cleaning head. The generated vacuum collects debris from a surface to be cleaned and deposits the debris, for example, in a debris collector. The vacuum may also include a motor to rotate a brushroll within the cleaning head. The rotation of the brushroll agitates debris that has adhered to the surface to be cleaned such that the generated vacuum is capable of removing the debris from the surface. In addition to electrical components for cleaning, the vacuum cleaner may include one or more light sources to illuminate an area to be cleaned.

(3) Portable surface cleaning devices, such as hand-held vacuums, are generally more convenient than “corded” vacuums that couple to AC mains. However, one drawback to portable vacuum cleaners is that their power source, e.g., one or more rechargeable battery cells, allow for relatively limited amounts of cleaning time before recharging is necessary. Accessories such as brushrolls increase cleaning performance in some applications such as the cleaning of carpeted surfaces, upholstery, etc., but motors that drive brushrolls can consume significant power during use, and in particular, when the brushrolls are under load such as is the case when cleaning thick carpets and other high-friction surfaces. Accordingly, some hand-held surface cleaning devices do not include nozzles with brushrolls, while others offer removable brushrolls that a user can remove or otherwise disable to extend battery life.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings included herewith are for illustrating various examples of articles, methods, and apparatuses of the teaching of the present specification and are not intended to limit the scope of what is taught in any way.

(2) FIG. **1** shows an example surface cleaning device that includes nozzle control circuitry consistent with embodiments of the present disclosure.

(3) FIG. **2** shows an example method for controlling brushroll speed in accordance with an embodiment of the present disclosure.

(4) FIG. **3**A shows an example approach to utilizing detected velocity to adjust brushroll rotations per minute (RPM) in accordance with an embodiment of the present disclosure.

(5) FIG. **3**B shows an example approach to utilizing acceleration and/or orientation to detect when a surface cleaning device traverses a wall or other vertical surface, in accordance with an embodiment of the present disclosure.

(6) FIG. **3**C shows another example approach to detecting the presence of a wall using acceleration data, in accordance with an embodiment of the present disclosure.

(7) FIG. **4** shows an example surface cleaning device implementing nozzle control circuitry

consistent with the present disclosure.
(8) FIGS. **5**A-**5**B show another example surface cleaning device implementing nozzle control circuitry consistent with the present disclosure.
(9) FIG. **6** shows an example circuit diagram for a transistor switching circuit suitable for use in the nozzle control circuitry of FIG. **1**.
(10) FIG. **7** shows an example schematic diagram of a microcontroller suitable for use in the nozzle controller circuitry of FIG. **1**.
DETAILED DESCRIPTION
(11) In general, the present disclosure is directed to nozzle control circuitry (also referred to herein as nozzle circuitry) for use in surface cleaning devices that preferably reduces overall power consumption of a surface cleaning device by detecting the start of a cleaning operation by a user before energizing one or more components such as an agitator. The nozzle control circuitry can detect a cleaning operation based on data output from one or more sensors (also referred to herein as operation sensors). For example, the nozzle control circuitry can communicate with at least one of a motion sensor such as an accelerometer, an orientation sensor such as gyroscope, and/or an air pressure sensor operatively coupled within a dirty air inlet to detect the presence of generated suction.
(12) Preferably, the nozzle control circuitry and one or more operation sensors are disposed or otherwise integrated into a removable nozzle housing (or attachment housing) such that the removable nozzle housing can be selectively coupled to a surface cleaning device and operate without necessarily communicating electrically and/or physically with other components of the surface cleaning device. The nozzle control circuitry also preferably detects the end of a cleaning operation using one or more of the aforementioned operation sensors and can de-energize power to one or more associated components such as an agitator without necessarily requiring user input, e.g., a button press.
(13) In more detail, the nozzle control circuitry preferably includes one or more power supplies (e.g., battery cell(s), and preferably rechargeable battery cells), a controller (also referred to herein as a nozzle controller or nozzle microcontroller), and operation sensor(s) collocated within/on a nozzle housing to implement a nozzle control scheme that can enable, adjust, and disable brushroll action and/or other nozzle-based components (e.g., LED status lights, side brushes, nozzle angle/height adjuster, and so on) without the necessity of receiving user input (e.g., button input) or having electrical communication between the nozzle control circuitry and the circuitry controlling the suction motor, for example.
(14) The nozzle control scheme preferably operates in a relatively low-power mode to detect usage of the surface cleaning device by momentarily powering a nozzle controller and one or more operation sensors (e.g., gyroscope, accelerometer, magnetometer, pressure sensor) to detect usage of a surface cleaning device by comparing sensor data to associated predefined thresholds, for example. Once usage/operation of the surface cleaning device is detected, the nozzle control circuitry can preferably transition to a relatively high-power, in-use mode to, for instance, drive brushroll motor(s) to vary RPMs of a brushroll/agitator based on a detected floor type, and select from predefined RPM values as the nozzle encounters different detected floor types. The nozzle control circuitry then preferably automatically transitions back to the low-power mode after the same detects cleaning operations have ended, e.g., based on sensor data measuring below predefined thresholds for a predefined period of time. The transition to the low-power mode preferably includes the nozzle control circuitry de-energizing the brushroll motor and/or related components and then returning to a sequence that can include momentarily energizing the nozzle controller and operation sensor(s), as discussed above, until subsequent usage is detected.
(15) Accordingly, nozzle control circuitry consistent with the present disclosure can preferably perform relatively low-power, coarse-grain sampling of sensor data and intelligently transition to a relatively high-power mode of operation, generally referred to herein as an in-use mode, that can

include brushroll action, activation of optional side brushes, enabling an LED to increase visibility within a surrounding environment, diagnostic output (e.g., battery charge level via LEDs), deployment of a cleaning solution, height adjustment of brushroll/agitator, and/or clog detection.
(16) Preferably, the nozzle control circuitry can perform relatively fine-grain detection of surface types while in the in-use mode to ensure optimal cleaning operation, e.g., through adjusting RPM of brushroll(s), deployment of a cleaning solution, and/or alerting a user to a detected clog condition. Stated differently, the nozzle control circuitry can adjust to surface-type changes in a timely fashion, e.g., within 1-3 seconds, and preferably within 1 second, to ensure optimal cleaning.
(17) In addition, the nozzle control circuitry is preferably implemented within a single nozzle housing thus eliminating the necessity of wires/interconnects to electrically couple the nozzle control circuitry to circuitry governing suction motor operation, for instance. This increases aesthetic appeal of a surface cleaning device by eliminating wires/interconnects, and allows for the nozzle control circuitry to operate in an independent fashion. In cases where the nozzle includes one or more batteries, overall battery life of a surface cleaning device can be extended as the nozzle components can draw power from the nozzle batteries rather than from a primary power source such as batteries that are configured to power a suction motor and associated circuitry.
(18) In one specific non-limiting example embodiment, a removable nozzle housing can further include an agitator motor and one or more brushrolls. The removable nozzle housing may then be coupled to a nozzle coupling section of a surface cleaning device when agitator-assisted cleaning is desired. The nozzle control circuitry within the removable nozzle housing may then detect the initiation of a cleaning operation and energize the agitator motor without requiring additional user input, e.g., a button press. More preferably, the removable nozzle housing includes an integrated power supply such as one or more battery cells to power the nozzle control circuitry and/or associated components such as the agitator motor and associated brushroll(s) to avoid adding load to the primary power source of the surface cleaning device, e.g., the power source used to power a suction motor. The removable nozzle housing may then be stored separate from the surface cleaning device, and in addition, optionally allow for charging of the integrated power supply within the removable nozzle housing via a dock or other suitable device such as a power adapter coupled to AC mains.
(19) As generally referred to herein, dust and debris refers to dirt, dust, water, and/or any other particle that may be pulled by suction into a surface cleaning device.
(20) Turning to the Figures, FIG. **1** illustrates an example surface cleaning device **101** that implements nozzle control circuitry **100** consistent with the present disclosure. The embodiment of FIG. **1** illustrates the surface cleaning device **101** as a hand-held surface cleaning device, and therefore, the following disclosure may also refer to the surface cleaning device **101** as a hand-held vacuum or hand-held surface cleaning device. However, other types of surface cleaning devices can implement aspects and embodiments of the nozzle control circuitry **100** disclosed herein such as so-called stick vacs, robotic vacuums, or any other device that seeks to intelligently engage brushrolls and/or adjust operating modes of a nozzle to optimize cleaning performance and/or extend battery life.
(21) As shown, the surface cleaning device **101** includes a handle portion (or handle) **114**, body (or base) **116**, and a nozzle **102**. The body **116** includes a removable dust cup **117** for receiving and storing dirt and debris. The body **116** can fluidly couple with the nozzle **102** to receive dirt and debris for storage within the removable dust cup **117**.
(22) The handle portion (or handle) **114** preferably includes a shape/profile contoured to a user's hand to reduce wrist fatigue during use. Preferably, one or more user interface buttons adjacent the handle portion **114** permit ON/OFF of a suction motor, and removal of the removable dust cup **117** for purposes of emptying dirt and debris, for example. The handle portion **114** transitions to the body **116**, with the body **116** providing a cavity to house the removable dust cup **117**, optional

filter, and vacuum controller circuitry **118**.

(23) Preferably, the vacuum controller circuitry **118** includes a primary controller **120**, suction motor **122**, and primary power supply **124**. Each component of the vacuum controller circuitry **118** resides within the body **116**, e.g., each component is collocated within the body **116**, although the components can reside in different locations, e.g., within the handle portion, additional housing sections, and so on, depending on a desired configuration. The primary controller **120** includes circuitry such as a microcontroller, application-specific integrated circuit (ASIC), and/or discrete circuitry, logic, memory and chips. Likewise, the primary controller **120** can perform methods as variously described herein using hardware (e.g., a processor, ASIC, circuitry), software (e.g., computer-readable code compiled or interpreted from assembly code, C++ code, C code, or an interpreted language such as Java), or any combination thereof.

(24) The primary controller **120** further includes circuitry to provide a driving signal to cause the suction motor **122** to turn ON/OFF and increase/decrease suction during cleaning operations. This circuitry can include, for instance, a motor driving circuit, power conversion circuitry, speed regulators, and so on. The suction motor **122** can comprise a DC motor or other suitable device for generating suction. In operation, the suction motor **122** can thus generate suction to draw air into an inlet of the nozzle **102**.

(25) The primary controller **120** and the suction motor **122** each draw power from the primary power supply **124**. The primary power supply **124** can include one or more battery cells, and preferably rechargeable battery cells such as rechargeable lithium ion battery cells, and associated circuitry such as DC-DC converters, voltage regulators and current limiters.

(26) Continuing on, the nozzle **102** is preferably configured to removably couple to the body **116**. The nozzle **102** preferably defines a dirty air passageway, and a dirty air inlet fluidly coupled to the dirty air passageway. Preferably, the nozzle **102** includes one or more brush rolls (not shown), and the nozzle control circuitry **100** disposed thereon, such as shown in FIG. **1**.

(27) The nozzle control circuitry **100** preferably includes a nozzle controller **104**, a nozzle power supply **106**, operation sensor(s) **108**, optional floor detect circuitry **110**, and a brushroll motor **112**. Preferably, the optional floor detect circuitry **110** comprises at least one floor type sensor (also referred to herein as a floor type detector).

(28) The nozzle controller **104**, which may also be referred to as a secondary controller, can be implemented as a microprocessor, ASIC, circuitry, software instructions, or any combination thereof. Note, while the nozzle controller **104** is shown as a separate and distinct component from that of the primary controller **120**, the nozzle controller **104** may be implemented whole, or in part, by the primary controller **120**.

(29) The nozzle power supply **106** can include one or more battery cells, and preferably one or more rechargeable battery cells and associated circuitry. The nozzle power supply **106** may also be referred to as secondary power supply. Preferably, the nozzle control circuitry **100** is electrically isolated from the vacuum controller circuitry **118**. In this preferred example, no direct (e.g., a wire or other interconnect) extends across/within the surface cleaning device **101** to provide electrical communication between the vacuum controller circuitry **118** and nozzle control circuitry **100**. Stated differently, the nozzle control circuitry **100** and the vacuum controller circuitry **118** can operate independent of each other and can utilize dedicated power supplies such that the primary power supply gets dedicated to the components of the nozzle. Likewise, the primary power supply **124** can primarily power the vacuum control components, to the exclusion of the nozzle control circuitry **100**.

(30) Preferably, each of the nozzle control circuitry **100** and the vacuum controller circuitry **118** therefore include separate and distinct power supplies, e.g., nozzle power supply **106** and the primary power supply **124**, respectively. The primary power supply **124** can include a different number and/or type of battery cells than the nozzle power supply **106**. For instance, space constraints of the nozzle **102** relative to the body **116** can result in more/larger capacity battery cells

implemented within the primary power supply **124** relative to the nozzle power supply **106**.
(31) Likewise, the nozzle control circuitry **100** and the vacuum controller circuitry **118** preferably include separate external electrical contacts (not shown) for charging purposes. Thus, the nozzle **102** can be optionally decoupled and charged separately from the surface cleaning device **101**, and the surface cleaning device **101** can continue to be used for cleaning operations that do not necessarily require brushrolls/nozzle features. Alternatively, or in addition, the surface cleaning device **101** can couple to a docking station (not shown) with electrical contacts/mating connectors for the nozzle **102** and the body **116** such that the primary power supply **124** and nozzle power supply **106** can be charged simultaneously and/or sequentially by the same charging circuit.
(32) In an embodiment, the nozzle control circuitry **100** and the vacuum controller circuitry **118** are electrically coupled, e.g., via wires or other electrical interconnect. Thus, in this embodiment, the vacuum controller circuitry **118** may utilize power from the nozzle power supply in addition to the primary power supply to increase operational time for cleaning, or the surface cleaning device **101** can include a single power supply, e.g., primary power supply **124**, such that both the nozzle control circuitry **100** and the vacuum controller circuitry **118** utilize the same power supply.
(33) The operation sensor(s) **108** can include one or more sensors disposed on/in the nozzle **102**. For example, the operation sensor(s) **108** can include any sensor capable of sensing user-supplied pressure/force such as a strain gauge or other force sensor configured to measure the force of the nozzle **102** against the surface to be cleaned **103**, and/or any sensor capable of detecting suction force. Preferably, the nozzle **102** can include an air pressure sensor disposed along a dirty air passageway defined by the same to detect suction generated by the suction motor **122** and output a proportional electrical signal. In any such cases, the nozzle controller **104** may then receive output (also referred to herein as output data) from the operation sensor(s) **108** to detect usage of the surface cleaning device **101**.
(34) Alternatively, or in addition, the operation sensor(s) **108** can include an accelerometer, gyroscope, and/or magnetometer. In this example, the operation sensor(s) **108** may therefore include a motion sensing arrangement. The operation sensor(s) **108** can therefore detect acceleration of the surface cleaning device **101**, the direction of that movement (along 2 or more axis such as X, Y and Z), and/or the orientation of the nozzle, e.g., roll, pitch and yaw to determine an angle/orientation a user holds the surface cleaning device **101** relative to the surface to be cleaned. The operation sensor(s) **108** may then output data such as one or more signals with values representing the acceleration and orientation data in real-time or on a periodic basis depending on a desired sample rate. For example, the operation sensor(s) **108** can output data that indicates the surface cleaning device **101** is angled substantially transverse relative to a surface to be cleaned **103** (e.g., as shown in FIG. **1**). The nozzle controller **104** may then detect usage of the surface cleaning device **101** based on such output data from the operation sensor(s) **108**.
(35) Preferably, at least one of the operation sensor(s) **108** operates in a low-power mode whereby the at least one operation sensor operating in the low-power mode can be used to detect threshold events before utilizing relatively high-power sensors, increasing sample rates from the operation sensor(s), and/or energizing nozzle components such as the brushroll motor **112**. For instance, the operation sensor(s) **108** can include a motion sensor (e.g., an accelerometer or other acceleration sensor) or motion sensor arrangement (e.g., gyroscope, accelerometer, and/or magnetometer) and periodically sample acceleration data in a low-resolution manner, e.g., once per second, to detect movement of the surface cleaning device **101** by a user during the low-power mode. In the event the sampled acceleration data exceeds a predefined threshold value over a period of time T, the nozzle controller **104** may then transition to a normal or “in-use” mode, whereby a driving signal is provided to the brushroll motor **112** to cause the same to rotate one or more brush rolls during cleaning, and/or another cleaning mode is activated such as side brush activation. Alternatively, or in addition, the driving signal may also cause cleaning fluid to be dispensed from a cleaning fluid reservoir (not shown) disposed in the body **116** or the nozzle **102**.

(36) Preferably, the in-use mode can also include activating one or more light sources disposed on the surface cleaning device **101** to increase visibility, e.g., one or more headlamp bulbs, LEDs, and/or power diagnostic lights such as LEDs that can indicate charge levels of the nozzle power supply **106** and/or the primary power supply **124**, error conditions, clogs restricting inlet airflow into nozzle **102**, filter change reminders, or other operating conditions.
(37) Alternatively, or in addition, pressure measurements from the operation sensor(s) **108** can be utilized to validate or otherwise detect the surface cleaning device **101** is in use by a user. For instance, the nozzle **102** being pressed against a surface to be cleaned can trigger a pressure/force measurement value output by the operation sensor(s) **108** that can be used alone or in combination with movement data by the nozzle controller **104** to transition to the in-use mode. For example, a force measurement that exceeds a first predefined threshold may then trigger a suction measurement by an air pressure sensor. In this example, the nozzle controller **104** may then transition to the in-use mode based on the suction measurement exceeding an associated predefined threshold. More preferably, the operation sensor(s) **108** include one or more sensors within or adjacent the dirty air inlet **105** of the nozzle **102** to detect suction levels, and based on those measured levels exceeding a threshold, the nozzle control circuitry **100** and more particularly the nozzle controller **104** can transition to the in-use mode.
(38) Therefore, in view of the foregoing, the nozzle control circuitry **100** preferably utilizes a low-power mode or “standby” mode whereby sampling is performed in a relatively coarse-grain fashion using low-resolution sampling or low-power sensors, or both, to reduce power consumption and extend battery life. The nozzle control circuitry **100** can then transition to the in-use mode during cleaning operations, e.g., to provide brushroll action, dispense cleaning fluid, provide illumination to aid in cleaning operations, and/or display operational status to a user, and then preferably detect the end of a cleaning operation based on pressure measurements and/or motion data (or a lack thereof) to transition back to the low-power mode, e.g., to conserve power. Thus, from a user's perspective the surface cleaning device **101** automatically detects that such nozzle features are desired by a user's natural motions, the presence of suction being generated by the suction motor **122**, and/or by the orientation the user is holding the surface cleaning device **101** relative to the surface to be cleaned **103** during cleaning operations, and when such nozzle features can be automatically disabled, e.g., to conserve power.
(39) One example method for transitioning between a standby/sleep mode for the nozzle control circuitry **100** and the in-use mode is as follows. Initially when the surface cleaning device moves, e.g., based on a user gripping the handle portion **114** and moving surface cleaning device **101**, an accelerometer or other motion sensor of the operation sensor(s) **108** sends a signal to a transistor switching circuit (not shown) of the nozzle control circuitry **100**. The transistor switching circuit then causes the nozzle controller **104** to momentarily energize. More preferably, the transistor switching circuit also causes a light or other light source of the surface cleaning device **101** to also energize and provide illumination during cleaning operations without necessarily energizing other components such as the brushroll motor **112**.
(40) Preferably, the energized nozzle controller **104** then receives output data from at least one air pressure sensor of the operation sensor(s) **108** to determine if the output value exceeds a predefined threshold, and thus, if the surface cleaning device **101** is ON and in use by a user. Stated differently, the energized nozzle controller **104** can utilize a pressure sensor within the nozzle **102** to detect the presence of suction generated by the suction motor **122**.
(41) Preferably, the nozzle **102** removably couples to the body **116** such that the nozzle **102** and operation sensor(s) **108** remain coupled together when the nozzle **102** is decoupled from the body **116** of the surface cleaning device **101**. More preferably, at least the nozzle **102**, the operation sensor(s) **108**, and the brushroll motor **112** remain coupled together when the nozzle **102** is decoupled from the body of the surface cleaning device **101**.
(42) In response to the energized nozzle controller **104** determining the surface cleaning device **101**

is ON, e.g., in use, the nozzle controller **104** then transitions to the in-use mode. The nozzle controller **104** can then optionally cause one or more of the headlamp(s), diagnostic LEDs, and brushroll motor **112** to switchably energize and turn ON. While ON, the nozzle controller **104** preferably periodically receives output data from the operation sensor(s) **108** to detect continued use via, for example, acceleration, orientation of the surface cleaning device, and/or measured pressure (e.g., suction). During the in-use mode, the nozzle controller **104** preferably utilizes a motion sensor of the operation sensor(s) **108**, such as an accelerometer, and floor detect circuitry **110** to control the brush motor mode/RPM relative to the detected floor type, as discussed in greater detail below.
(43) Preferably, the nozzle controller **104** samples output data of the pressure sensor of the operation sensor(s) **108** to determine if the output data indicates the surface cleaning device **101** remains in use, e.g., based on comparing the output data to a predetermined threshold. More preferably, the nozzle controller **104** preferably continuously samples the pressure sensor, e.g., every 50 ms to 1000 ms, and preferably every 500 ms, to determine if the current pressure level indicates the surface cleaning device **101** is in use. Thus, in response to air pressure values and/or acceleration measurements falling below the predefined threshold for a predefined period of time (e.g., 1-20 seconds, and preferably 2-3 seconds), the transistor switching circuit of the nozzle control circuitry **100** switches “low” to turn off/de-energize the nozzle controller **104**, the brushroll motor **112**, and/or other components of the nozzle control circuitry **100** such as the floor detect circuitry, to transition the same to the low-power/standby mode.
(44) In an embodiment, the nozzle controller **104** can use output data from the operation sensor(s) **108** and floor detect circuitry **110** to control the brush motor RPMs. One such example method for dynamically controlling the brush motor mode/RPM based on a detected floor type is discussed further below with reference to FIG. **2**.
(45) Smart Nozzle Architecture and Methodologies
(46) Turning briefly to FIGS. **6** and **7** with reference to FIG. **1**, FIG. **6** shows an example circuit diagram for a transistor switching circuit **600** suitable for use in the nozzle control circuitry of FIG. **1**, and FIG. **7** shows an example schematic diagram of a microcontroller **700** (MCU) suitable for use in the nozzle controller circuitry of FIG. **1**, and preferably, suitable for use as the nozzle controller **104**.
(47) One example method for transitioning between a standby/sleep mode for the nozzle circuitry **100** and the in-use mode is as follows. Initially when the vacuum moves, e.g., based on a user gripping the handle portion **114** and moving the surface cleaning device**101**, an accelerometer of the operation sensor(s) **108** sends a signal to the transistor switching circuit **600** (See FIG. **6**) to briefly power a microcontroller **700**.
(48) The MCU **700** then receives an output value from a pressure sensor of the operation sensor(s) **108** to determine if the output value exceeds a predefined threshold, and thus, if the surface cleaning device **101** is ON and in use by a user. In response to the MCU **700** determining the surface cleaning device **101** is ON, the transistor switching circuit **600** remains high keeping the MCU **700** powered on to transition to the in-use mode. The MCU **700** can then cause one or more of the headlamp(s), diagnostic LEDs, and brushroll motor **112** to turn ON. While ON, the MCU **700** periodically receives output data from the pressure sensor of the operation sensor(s) **108** to confirm values that indicate pressure consistent with usage of the surface cleaning device **101**. During the in-use mode, the MCU **700** can utilize an accelerometer of the operation sensor(s) **108**, floor detect circuitry (e.g., implemented in the pressure sensor and/or other suitable sensor in combination with a floor detect algorithm) to intelligently control the brush motor mode/RPM relative to the detected floor type.
(49) In response to pressure values falling below the predefined threshold for a predefined period of time (e.g., between 1-20 seconds, and preferably between 1-3 seconds), the transistor switching circuit **600** switches “low” to turn off the MCU **700** and transition to the low-power, standby mode.

(50) In an embodiment, the surface cleaning device **101** initially provides a signal_A to the transistor switching circuit **600** to momentarily enable the MCU **700** (e.g., to wake from sleep/standby mode). The MCU **700** then can sets different signal_B that is OR′d with signal_A, to keep the MCU **700** powered.
(51) The MCU then can sample the operation sensor(s) **108** to determine if the output value indicates the surface cleaning device **101** is ON (e.g., based on a threshold value). If the surface cleaning device **101** is ON, then MCU **700** keeps signal_B on, else it makes signal_B off, therefore transitioning the smart nozzle circuitry back into sleep/standby mode. If the surface cleaning device **101** is detected as ON, then the MCU **700** then tells the headlamps, debug led, and brushroll motor **112** to turn on, for example.
(52) Once the MCU **700** is ON and initialized, the MCU **700** continuously samples the pressure sensor of the operational sensor(s) **108** to determine if the current pressure level indicates the surface cleaning device **101** is ON (e.g., based on a threshold), and if not, then the MCU **700** it sets signal_B off, thereby transitioning the mode to standby.
(53) Additionally, the MCU **700** can use motion sensor data, floor detect circuitry (e.g., established using floor detect algorithm, as discussed above) and the pressure sensor measurements to intelligently control the brush motor mode/RPM. One such example method for intelligently controlling the brush motor mode/RPM is discussed with reference to FIG. **2**.
(54) The components of the nozzle control circuitry **100** can be disposed on a single substrate, e.g., a printed circuit board (not shown), and be powered by the nozzle power supply **106** implemented as a 16V lithium Ion battery, for instance. In this case, the 16V output can be fed through a DC-DC converter circuit (not shown) to step down to 12V to power, for instance, the brushroll motor **112**. The 12V output of the DC-DC converter circuit can be then fed through another DC-DC converter circuit (not shown) to step down the voltage to a (constant) 3.3V source to supply power to the sensory, such as the operation sensor(s) **108**, floor detect circuitry **110**, and diagnostic LEDs, for example.
(55) Consistent with the present disclosure, the aforementioned in-use mode can include additional operational features. In an embodiment, the nozzle controller **104** receives acceleration data from the operation sensor(s) **108**. When movement in a negative X or Y direction (e.g., indicating the user is pulling the surface cleaning device **101** towards them), exceeds a predefined threshold, the nozzle controller **104** can de-energize the brushroll motor **112** to advantageously reduce strain experienced by the by the user when pulling the surface cleaning device **101** backwards during cleaning operations.
(56) In-Use Detection via Air Pressure Sensor(s)
(57) In an embodiment, the operation sensor(s) **108** include at least one air pressure sensor that gets turned ON (e.g., energized) after the same receives a signal from the nozzle controller **104**. The nozzle controller **104** may provide the signal to turn on the at least on air pressure sensor based on, for example, acceleration data received from an accelerometer of the operation sensor(s) **108**.
(58) The at least one air pressure sensor can then measure/read pressure values within the surface cleaning device **101** and communicate the pressure values to the nozzle controller **104**. When the pressure values are below a first predefined pressure threshold value (or a minimum pressure value) that indicates the surface cleaning device **101**, and in particular the suction motor **122**, is OFF, the nozzle controller **104** then causes the brushroll motor **112** to de-energize and turn OFF. On the other hand, if the pressure value is above a second predefined pressure threshold value (or a maximum pressure value), the nozzle controller **104** can determine/detect that the surface cleaning device **101** is ON/in-use. Thus, the nozzle controller **104** can determine that the surface cleaning device **101** is in use (or not, as the case may be) via a plurality of different threshold pressure values.
(59) In an embodiment, the nozzle controller **104** communicates with the pressure sensor at a relatively high-frequency so that the nozzle controller **104** can monitor that the surface cleaning

device **101** remains ON/in-use and initiate timely transition between the in-use and standby power modes. In this embodiment, the pressure sensor of the operation sensor(s) **108** may be used exclusively, or in combination with other sensors of the operation sensor(s) **108**, e.g., an accelerometer, to identify whether the surface cleaning device **101** is an ON (e.g., in-use) or in an OFF (e.g., storage/standby) mode. The speed (RPM) at which the brushroll motor **112** operates the brushroll(s) can be controlled via the nozzle controller **104**, or preferably, based on the floor detect circuitry **110**.

(60) The following hand-held surface cleaning device states (modes) may be detected by the nozzle controller **104**, and operation of the nozzle **102** may be adjusted accordingly. The nozzle controller **104** preferably detects a high suction mode (or bare floor mode), when a pressure sensor of the operation sensor(s) **108** indicates a pressure value above a predefined threshold and/or when the floor detect circuitry **110** detects presence of a bare floor. In this mode, the nozzle controller **104** may adjust the RPM of the brushroll(s) preferably to zero RPMs.

(61) Conversely, the nozzle controller **104** preferably detects a low suction mode, or carpet mode, based on the pressure sensor indicating a pressure value below the predefined threshold and/or the floor detect circuitry **110** detects the presence of carpet. In this mode, the nozzle controller **104** may then adjust the RPM of the brushroll(s), and preferably increase RPMs to a predetermined rate relative to the bare floor mode. Note, the predefined threshold for the high suction mode (or bare floor mode) and the low suction mode (or carpet mode) may be the same, or different, depending on a desired configuration.

(62) FIG. **2** shows an example method **200** for detecting a floor type by monitoring current across the brushroll motor **112**. The monitoring/measurements of the current drawn by the brushroll motor **112** may be performed by the nozzle controller **104** or other suitable circuitry, and preferably, circuitry disposed within the nozzle **102**. When the surface cleaning device **101** transitions to the in-use mode, the nozzle controller **104** can preferably utilize an on-board ADC or other suitable circuitry and amplify and convert a measured electrical current drawn by the brushroll motor **112** into a proportional voltage. Then, the amplified and converted voltage can be provided to the nozzle controller **104**. The nozzle controller **104** may then monitor the amplified and converted voltage via method **200**. The nozzle controller **104** can be configured to execute the method of FIG. **2**, although other components can perform one or more acts of method **200**.

(63) In act **202**, the nozzle controller **104** detects the surface cleaning device **101** is in use. As discussed above, usage of the surface cleaning device **101** can be determined by detecting that the suction motor **122** is generating suction and/or via acceleration data, for example. The nozzle control circuitry **100** can thus transition to the in-use mode based on detecting usage of the surface cleaning device **101**. Other approaches to detecting usage of the surface cleaning device **101** are also applicable and this disclosure is not intended to be limited in this regard. For instance, non-limiting alternatives include wireless communication (WIFI™, BLUETOOTH™ low energy, NFC) between the nozzle control circuitry and the vacuum controller circuitry **118**, vibration measurements and/or sound measurements.

(64) In act **204** the nozzle controller **104** sets the current mode for the nozzle to FLOOR mode (also referred to herein as a bare floor mode). FLOOR mode includes an associated RPM, which the nozzle controller **104** can determine via a look-up table in a memory, for example. The nozzle controller **104** therefore sets the current mode to the FLOOR mode by driving the brushroll motor **112** to rotate at the associated RPM. In an embodiment, the FLOOR mode is between 0 and 100% of potential RPM speed, and preferably, zero (0) RPMs.

(65) In act **206**, a first measurement timer is set. In act **208**, the nozzle controller **104** receives a plurality of electrical current measurements for the period of time defined by the first measurement timer. By way of example, the first measurement timer may be set to 1200 milliseconds. In the event the timer elapses, the method **200** can transition the nozzle from FLOOR mode to OFF and optionally return to act **202**.

(66) In act **208**, the nozzle controller **104** receives a plurality of electrical current measurement values. For instance, the nozzle controller **104** can receive up to at least five (5) measurement values by sampling at a rate of 40 ms. Thus, at 200 ms, the nozzle controller **104** can have 5 current measurement values in this example, although other sampling rates are within the scope of this disclosure. Preferably, the sampling rate is at least 40 ms, and more preferably, at least 100 ms.
(67) In act **210**, the nozzle controller **104** averages the received plurality of current measurements to produce a first current average (AVG**1**). In act **212**, the nozzle controller **104** determines if the first current average (AVG**1**) exceeds a first predefined threshold. If the first current average (AVG**1**) exceeds the first predefined threshold, the method **200** continues to act **214**, if not, the method **200** returns to act **204** and continues to perform acts **204-212**.
(68) In act **214**, the nozzle controller **104** transitions the mode from FLOOR MODE to CARPET MODE. Transitioning to the CARPET MODE can further include the nozzle controller **104** driving the brushroll motor **112** at an associated RPM, the associated RPM of the CARPET MODE being greater than the associated RPM of the FLOOR MODE.
(69) In act **218**, the first measurement timer is optionally cancelled (or disabled), and a second measurement timer is set. The duration of the second measurement timer may be less than that of the duration of the first measurement timer. For instance, the second measurement timer may be set to 700 ms, or another value. Preferably, the second measurement timer is 500 ms or less.
(70) In act **220**, the nozzle controller **104** samples the electrical current drawn by the brushroll motor **112** every X ms, e.g., 40 ms or less. In act **222**, the nozzle controller **104** averages the current measurement values to determine a second current average (AVG**2**). In act **224**, the nozzle controller **104** determines if the second current average (AVG**2**) is less than a predefined threshold, and if so, the method continues to act **226**. Otherwise, the method **200** returns to act **220** and continues to perform acts **220-224**. In act **226**, the nozzle controller **104** transitions the mode from carpet mode to floor mode, and the method **200** then continues to act **204**.
(71) Thus, nozzle control circuitry is disclosed herein that can include a separate battery from an associated hand-held vacuum and can be independently powered and operated from the hand-held vacuum to eliminate wires/interconnects extending through the hand-held vacuum to the nozzle. Preferably, a pressure sensor is used to determine the operation mode of the surface cleaning device based on detecting the presence of suction generated by a suction motor.
(72) Preferably, the nozzle controller **104** uses acceleration data to determine the forward/backward motion of the nozzle **102**. When detecting backward motion, the speed of the brushroll may be reduced or increased by the nozzle controller **104** to reduce drag friction that causes tiring of user arms. Alternatively, or in addition, the direction of rotation of the brush roll(s) may be changed such that the brushrolls(s) “pull” the surface cleaning device **101** in a direction generally corresponding to the direction of travel desired by the user. Preferably, output data from an accelerometer can also be used to determine the forward/backward motion of the nozzle **102**, and the nozzle controller **104** will preferably conserve battery run-time by reducing the nozzle speed based on direction of motion (e.g. in the back stroke). In addition, the nozzle controller **104** can utilize a pressure sensor to determine a clog in the system and can alert the user to service the clog. Such clog determination can be based on measured pressure vs a look-up of expected pressure.
(73) FIGS. **3**A-**3**C show additional aspects of a nozzle consistent with the present disclosure. As shown, the accelerometer data can be used to identify a “backstroke” whereby a user pulls the nozzle towards themselves, such as shown in FIG. **3**A. In response, the nozzle can reduce brushroll speed to reduce friction introduced by the same to decrease user fatigue. Also, accelerometer/gyro data can be utilized to detect when a nozzle traverses a vertical or substantially vertical surface such as a wall, such as show in FIG. **3**B. In addition, and as shown in FIG. **3**C, a nozzle consistent with the present disclosure can detect contact with a wall, e.g., based on sudden deceleration, and may modify brushroll and/or wheel speed to reduce the amount of user force necessary to draw the nozzle away from the wall to continue cleaning operations.

(74) FIG. **4** shows an example surface cleaning device **400** implementing nozzle control circuitry consistent with the present disclosure. As shown, the example surface cleaning device **400** includes a body **402** coupled to a nozzle **406** via a wand **404**. The nozzle **406** can implement the nozzle control circuitry **100** as discussed above.
(75) FIGS. **5**A-**5**B show another example surface cleaning device **500** implementing nozzle control circuitry consistent with the present disclosure. As shown, the example surface cleaning device **500** includes a wandvac **502** that removably couples to a nozzle **504**. The nozzle **504** can implement the nozzle control circuitry **100** as discussed above.
(76) In one preferred example, accelerometer data may be used to determine if the nozzle **102** is cleaning near a wall by sensing "bumping" and may cause change to the brushroll speed or cause a different side brush motor to turn on to optimize side cleaning.
(77) Table 1 shows various user operations using nozzle control circuitry consistent with the present disclosure and the resulting action and intended benefits.
(78) TABLE-US-00001 TABLE 1 User Trigger/ Operation Stimuli Operational Result Benefit Backstroke Velocity Change: Less resistance pull- Brush roll speed; force on operators Suction force; hand. Light Intensity; Reduced energy Articulate rear consumption bristle strip; and/or resulting in front edge shutter increased run-time (particularly important for cordless products) Communicate Intelligent Behavior and energy conservation. Improved cleaning on backstroke. Traversing Velocity Articulated side Better edge a wall/ brushes/side edges cleaning. object Airflow Directivity Communicate Change: Intelligent Behavior Brush roll speed; and Energy Suction force; Conservation. and/or Light intensity Hitting a Acceleration/ Change: Better front edge baseboard Deceleration Brush roll speed; cleaning. (front strike) Suction Force; and/or Articulate Frontedge Shutter. Surface type Acceleration/ Change: Optimized detection Deceleration Brushroll speed; speed/suction for and/or better cleaning on Suction force. any floor type. User Acceleration/ Change: Customized interaction Deceleration Brush roll speed; speed/suction for style and/or better cleaning. detection Suction force.
(79) In accordance with an aspect of the present disclosure a surface cleaning device is disclosed. The surface cleaning device comprising a body defining a handle portion and a dirty air passageway, a suction motor for generating suction to draw air into the dirty air passageway, a nozzle coupled to the body and having a dirty air inlet fluidly coupled with the dirty air passageway, a sensor coupled to the nozzle, a brushroll motor to drive one or more brush rolls, and nozzle control circuitry, the nozzle control circuitry to detect usage of the surface cleaning device based on output data from the sensor and, in response to receiving the output data, cause the brushroll motor to energize.
(80) In accordance with another aspect of the present disclosure a hand-held surface cleaning device is disclosed. The hand-held surface cleaning device comprising a body defining a handle portion and a dirty air passageway, a suction motor for generating suction to draw dirt and debris into the dirty air passageway, a nozzle coupled to the body and having a dirty air inlet fluidly coupled with the dirty air passageway, the nozzle defining a cavity, a brushroll motor to drive one or more brush rolls within the cavity of the nozzle, and nozzle control circuitry disposed in the cavity of the nozzle, the nozzle control circuitry to detect usage of the hand-held surface cleaning device during a cleaning operation, and in response to detecting the usage of the hand-held surface cleaning device, sending a driving signal to the brushroll motor to cause the brushroll motor to rotate the one or more brush rolls at a predetermined rotations per minute (RPM).
(81) In accordance with another aspect of the present disclosure a method for controlling brushroll speed within a surface cleaning device is disclosed. The method comprising detecting, by a controller, a suction motor is generating suction to draw dirt and debris into an inlet of the surface cleaning device, in response to detecting suction generated by the suction motor, energizing a portion of a nozzle control circuit for detecting a floor type adjacent the inlet of the surface cleaning device, and sending a driving signal to a brushroll motor to adjust rotations per minute

(RPM) of one or more associated brush rolls based on the detected floor type.
(82) While the principles of the disclosure have been described herein, it is to be understood by those skilled in the art that this description is made only by way of example and not as a limitation as to the scope of the disclosure. Other embodiments are contemplated within the scope of the present disclosure in addition to the exemplary embodiments shown and described herein. It will be appreciated by a person skilled in the art that a surface cleaning apparatus may embody any one or more of the features contained herein and that the features may be used in any particular combination or sub-combination. Modifications and substitutions by one of ordinary skill in the art are considered to be within the scope of the present disclosure, which is not to be limited except by the claims.

Claims

1. A surface cleaning device comprising: a body defining a handle portion, a dirty air passageway, and a cavity to receive vacuum controller circuitry, the vacuum controller circuitry including: a primary controller; a primary power supply; and a suction motor for generating suction to draw air into the dirty air passageway; a nozzle removably coupled to the body and having a dirty air inlet configured to fluidly couple with the dirty air passageway, the nozzle including nozzle control circuitry that is electrically isolated from the vacuum controller circuitry, the nozzle control circuitry having: a brushroll motor to drive one or more brush rolls; an accelerometer configured to generate a motion signal corresponding to motion data in response to movement of the nozzle; a pressure sensor configured to generate a pressure signal corresponding to pressure data in response to pressure within the nozzle; a secondary power supply; and a secondary controller configured to: detect initial usage of the surface cleaning device prior to activation of the suction motor and the brushroll motor based, at least in part, on the motion signal corresponding to motion data; in response to detecting the initial usage and prior to activation of the suction motor and the brushroll motor, begin monitoring the pressure signal corresponding to pressure data; after detecting the initial usage, detect activation of the suction motor based, at least in part, on the pressure signal corresponding to pressure data indicating a change in pressure; and in response to detecting the activation of the suction motor, send a driving signal to the brushroll motor, the driving signal causing the brushroll motor to rotate the one or more brush rolls.
2. The surface cleaning device of claim 1, wherein the nozzle control circuitry de-energizes the secondary controller based on the pressure signal corresponding to pressure data indicating an amount of detected suction is below a predefined threshold.
3. The surface cleaning device of claim 1 further comprising at least one of: a force sensor to detect an amount of force supplied by the nozzle against a surface to be cleaned; or an orientation sensor to detect an orientation of the surface cleaning device relative to the surface to be cleaned.
4. The surface cleaning device of claim 1, wherein the nozzle control circuitry causes the one or more brush rolls to rotate at a predetermined rotations per minute (RPM).
5. The surface cleaning device of claim 4, wherein the surface cleaning device further comprises a floor type sensor and the nozzle control circuitry causes the brushroll motor to drive the one or more brush rolls at the predetermined RPM based on output from the floor type sensor.
6. The surface cleaning device of claim 1, wherein the nozzle control circuitry causes the brushroll motor to drive the one or more brush rolls at a predetermined rotations per minute based on the motion signal corresponding to motion data generated by the accelerometer.
7. A hand-held surface cleaning device comprising: a body defining a handle portion, a dirty air passageway, and a cavity to receive vacuum controller circuitry, the vacuum controller circuitry including: a primary controller; a primary power supply; and a suction motor for generating suction to draw dirt and debris into the dirty air passageway; and a nozzle removably coupled to the body and having a dirty air inlet configured to fluidly couple with the dirty air passageway, the nozzle

including nozzle control circuitry that is electrically isolated from the vacuum controller circuitry, the nozzle control circuitry having: a brushroll motor to drive one or more brush rolls; an accelerometer configured to generate a motion signal corresponding to motion data in response to movement of the nozzle; a pressure sensor configured to generate a pressure signal corresponding to pressure data in response to pressure within the nozzle; a light source; a secondary power supply; and a secondary controller configured to: detect initial usage of the hand-held surface cleaning device prior to activation of the suction motor and the brushroll motor based, at least in part, on the motion signal corresponding to motion data; in response to detecting the initial usage and prior to activation of the suction motor and the brushroll motor, begin monitoring the pressure signal corresponding to pressure data and energize the light source; after detecting the initial usage, detect activation of the suction motor based, at least in part, on the pressure signal corresponding to pressure data indicating a change in pressure; and in response to detecting the activation of the suction motor, send a driving signal to the brushroll motor to cause the brushroll motor to rotate the one or more brush rolls at a predetermined rotations per minute (RPM).

8. The hand-held surface cleaning device of claim 7 further comprising at least one of: a force sensor to detect a force value indicating contact between the nozzle and a surface to be cleaned; or an orientation sensor to detect an orientation of the hand-held surface cleaning device relative to the surface to be cleaned.

9. The hand-held surface cleaning device of claim 7, wherein the secondary controller causes the brushroll motor to change RPM of the one or more brush rolls based on the motion data.

10. The hand-held surface cleaning device of claim 7, wherein the nozzle control circuitry includes a floor type detector and the nozzle control circuitry causes the brushroll motor to adjust RPM of the one or more brush rolls based on output from the floor type detector.

11. A surface cleaning device comprising: a body including a dirty air passageway and a cavity to receive vacuum controller circuitry, the vacuum controller circuitry including: a primary controller; a primary power supply; and a suction motor for generating suction to draw air into the dirty air passageway; a nozzle configured to removably couple to the body and having a dirty air inlet configured to fluidly couple with the dirty air passageway, the nozzle including nozzle control circuitry that is electrically isolated from the vacuum controller circuitry, the nozzle control circuitry including: a brushroll motor to drive one or more brush rolls; an accelerometer configured to generate a motion signal corresponding to motion data in response to movement of the nozzle; a pressure sensor configured to generate a pressure signal corresponding to pressure data that is indicative of an air pressure within the nozzle; a secondary power supply; and a secondary controller configured to transition between a low-power mode and an in-use mode based, at least in part, on the motion signal corresponding to motion data, wherein: when in the low-power mode, the suction motor and the brushroll motor are inactive, and the secondary controller is configured to: monitor the motion signal corresponding to motion data to detect motion of the nozzle that corresponds to initial usage; and in response to detecting the initial usage, transition to the in-use mode; and when in the in-use mode the secondary controller is configured to: monitor the pressure signal corresponding to pressure data prior activating the brushroll motor; detect activation of the suction motor prior to activating the brushroll motor based, at least in part, on the pressure signal corresponding to pressure data indicating a change in pressure; and in response to detecting the activation of the suction motor, activating the brushroll motor such that the one or more brushrolls begin rotating.

12. The surface cleaning device of claim 11, wherein the motion signal corresponding to motion data is monitored at a first sample rate and the pressure signal corresponding to pressure data is monitored at a second sample rate, the second sample rate being greater than the first sample rate.

13. The surface cleaning device of claim 11, wherein, when in the low-power mode, the motion signal corresponding to motion data is monitored without the pressure signal corresponding to pressure data being monitored.